

NutriLens is an interactive Streamlit MVP built to test one core assumption: will budget-conscious users use a tool that recommends meals based on budget and nutrition goal before they order? This document walks through the current working prototype as it behaves today. Each section shows an implemented interaction, what it does, and where relevant, the validation insight it responds to.

01 Set your budget and nutrition goal

Preferences

Budget

RM12

Goal

High Protein

☐ Avoid poultry (prototype)

Prototype meal metadata only. This is not an allergy-safety check; unknown dishes may still contain poultry.

The sidebar is where every recommendation starts. A user selects one of eight budget tiers from RM6 to RM25 and chooses a nutrition goal: High Protein, Low Calorie, Best Value, or Balanced. Optional controls allow the user to search for a known meal, apply the prototype poultry filter, or preview an uploaded meal or menu image. In this walkthrough, the user sets RM12 and High Protein.

02 View the Best Match and alternatives

Your Best Match



Grilled Fish with Rice
RM12.00 | 490 kcal | 38g protein
✓ Within your RM12 budget
✓ Provides 38g protein
✓ Strong protein match
✓ Best match for High Protein

68/100

Other options



Grilled Chicken Rice
RM10.50 | 520 kcal | 35g protein
> Why this meal?

66/100



Chicken Rice
RM9.00 | 450 kcal | 28g protein
> Why this meal?

59/100



Chicken Satay (10 sticks)
RM10.00 | 450 kcal | 30g protein
> Why this meal?

59/100

Match Score reflects compatibility with your selected goal among meals within your budget — it is not a health or medical rating.

NutriLens first removes meals above the selected budget, then ranks the eligible options using a Match Score from 0-100 for the chosen nutrition goal. The highest-scoring meal is presented as the Best Match, followed by up to three alternatives. In this example, Grilled Fish with Rice is the Best Match at 68/100.

Match Score represents relative compatibility within the eligible options; it is not a health or medical rating.

03 Understand why a meal is recommended

Other options



Grilled Chicken Rice

RM10.50 | 520 kcal | 35g protein

Why this meal?

✓ Within your RM12 budget

✓ Provides 35g protein

✓ Strong protein match

66

/100

NutriLens explains the reasoning behind its recommendations instead of showing only a score. Alternative meals include an expandable ‘Why this meal?’ section that links the recommendation back to measurable factors such as budget, protein content, and the selected nutrition goal.

04 Search and evaluate a specific meal

Your Meal Recommendations

Selected Meal



Nasi Lemak

RM8.50 | 560 kcal | 20g protein | 65g carbs | 22g fat

✓ Within your RM12 budget

✓ Provides 20g protein

✓ Strong protein match

38

/100

Here's how it compares to your best options within budget:

Best Match



Grilled Fish with Rice

RM12.00 | 490 kcal | 38g protein

✓ Within your RM12 budget

✓ Provides 38g protein

✓ Strong protein match

✓ Best match for High Protein

68

/100

Users who already know what they are considering can search for a meal by name instead of relying only on the ranked list. Searching for Nasi Lemak displays its full nutrition profile and evaluates it against the current budget and goal. The Best Match remains visible underneath, allowing the user to compare the meal they intended to order with a stronger-ranked alternative.

Validation-driven refinement: manual search was retained as a lower-friction pathway for users who preferred looking up a known dish instead of scanning or uploading an image.

05 Apply the prototype poultry filter

Preferences

Budget

RM12

Goal

High Protein

✓ Avoid poultry (prototype)

Prototype meal metadata only. This is not an allergy-safety check; unknown dishes may still contain poultry.

Your Best Match

Meals known to contain poultry are hidden. Remaining ingredient status may be unverified.

Best Match



Grilled Fish with Rice

RM12.00 | 490 kcal | 38g protein

⚠ Poultry status unverified

✓ Within your RM12 budget

✓ Provides 38g protein

✓ Strong protein match

✓ Best match for High Protein

68

/100

Other options



Tuna Sandwich Set

RM8.00 | 350 kcal | 22g protein

⚠ Poultry status unverified

Why this meal?

53

/100

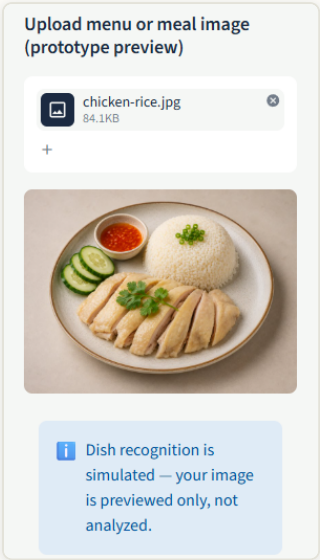
Users who want to avoid poultry can enable the prototype poultry filter. Meals known to contain poultry are removed before ranking. Meals whose poultry status is unknown remain visible but are clearly marked ‘Poultry status unverified’ rather than implied safe. The feature is intentionally limited to poultry metadata and is not an allergy-safety or ingredient-verification system.

Validation-driven refinement: this narrow filter was added after dietary-avoidance feedback and deliberately kept limited in scope for the MVP.

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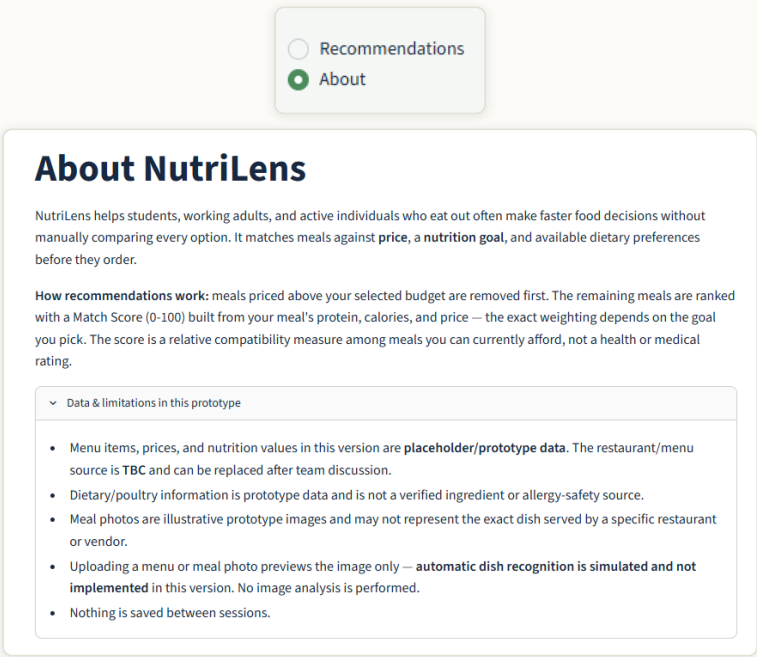
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06 Preview an uploaded meal or menu image



A user can upload a meal or menu image and immediately preview it inside NutriLens. In the current MVP, no recognition or analysis is performed on the uploaded image. Uploading an image does not change the recommendation results; recommendations continue to use the selected budget and nutrition goal. The interface states this boundary directly so prototype behavior is not mistaken for implemented AI recognition.

07 Review how NutriLens works and its prototype limitations



The About view explains the recommendation approach and consolidates the prototype’s limitations in one place. Current nutrition and meal information comes from prototype data rather than a live restaurant source. Poultry metadata is not verified ingredient data, meal photos are illustrative rather than verified vendor photography, image upload performs no recognition or analysis, and user activity is not saved between sessions.

PROTOTYPE SCOPE

The current NutriLens MVP uses a static prototype dataset of 15 meals and does not yet connect to live restaurant menus, user accounts, or persistent history. Match Score represents relative compatibility with the selected goal among eligible meals; it is not a health or medical rating. The poultry filter relies on prototype metadata rather than verified ingredient or allergy information, and uploaded images are previewed only. These boundaries are intentional: the MVP is designed to test the usefulness of budget- and nutrition-based meal recommendations before investing in full image recognition, restaurant integration, and persistent user features.